I2C VEML3328 Color Sensor

DEVLAB FACTOR

Mfr. Part #: UE0128

Product Reference Manual (V1.0)

Description

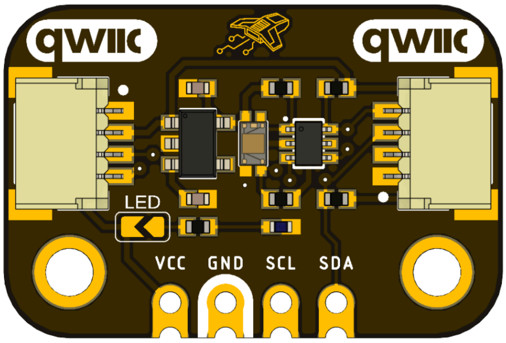
The **I2C VEML3328 Color Sensor** is a high-performance digital color sensing module designed for accurate ambient light and color measurement in embedded systems. The module integrates the **VEML3328** RGBIR (Red, Green, Blue, and Infrared) color sensor, providing high-resolution digital measurements through a standard **I²C communication interface**. The device combines photodiodes, low-noise analog front-end circuitry, integrated analog-to-digital converters, and programmable gain and integration time settings into a compact solution suitable for a wide range of color detection applications.

The sensor converts incident light into calibrated digital RGB and infrared values, allowing embedded systems to perform color recognition, ambient light compensation, white balance estimation, and illumination monitoring without requiring external analog signal conditioning.

DevLab format compatibility.

Simplicity and compatibility are the primary objectives of the DevLab form factor. It offers a board layout that is compact and optimized for serial communication, which includes I²C interface.

This format enables the establishment of rapid and dependable connections via a Protocol I²C connector that is entirely compliant with the Qwiic and STEMMA standards. This design guarantees efficient prototyping, accessibility, and seamless integration with both **external devices** and **the DevLab ecosystem**.



Key Features

- Integrated color sensor and signal conditioning IC
- Independent 16-bit measurements for Red, Green, Blue, Clear, and IR channels
- Analog Gain Ranges: ½ to 4 times
- Digital Gain Ranges: 1 to 4 times
- Integration Time Ranges: 50 ms to 400 ms
- Analog and Digital independent internal circuits
- Ultra-low power consumption
- Digital interfaces: I²C (Fast-mode Plus)
- Small form factor, RoHS compliant

Hardware Features

- Integrated RGBCIR color sensor and signal conditioning IC
- Independent 16-bit measurements for Red, Green, Blue, Clear, and IR channels
- Selectable digital gain ranges: x1(default), x2, x4
- Selectable analog gain ranges: x½, x1(default), x2, x4
- Digital communication via I²C through QWIIC and castellated holes in the board
- Hardware time synchronization between sensor and host
- Integrated power management unit for low-power operation

Software Support

The module is compatible with multiple embedded software environments and microcontroller platforms supporting I²C communication.

Supported environments include:

- Arduino IDE
- PlatformIO
- ESP-IDF
- MicroPython
- CircuitPython
- Raspberry Pi Pico SDK
- STM32 HAL / LL
- Zephyr RTOS

The device may be accessed using standard I²C transactions for configuration, light acquisition, and power-saving control.

Reference examples provided by UNIT Electronics are primarily focused on Arduino-based validation and evaluation applications. Additional platform compatibility may require software adaptation depending on the selected framework and system architecture.

Applications

- White balancing and color cast correction in digital cameras
- Automatic LCD backlight adjustment
- On/Off Light Switching in industrial and consumer applications
- Monitoring of LED Color output for IoT and smart Lighting’

CONTENTS

1 The Board

1.1 Accessories

- 1×4-pin 2.54 mm male header
- 1x QWIIC 4-pin JST 1.0 mm Harness

2 Ratings

2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
VCC	Input Supply Voltage	3.3	5.0	5.5	V
VDD	Internal Sensor Supply Voltage	3.2	3.3	3.4	V
f(SCL)	I²C Clock Frequency	10	—	400	kHz
Top	Operating Temperature	-25	25	+85	°C
t(BUF)	Bus Free Time Between START and STOP	1.3	—	—	µs
t(HD;STA)	Hold Time After Repeated START	0.6	—	—	µs
t(SU;STA)	Repeated START Setup Time	0.6	—	—	µs
t(SU;STO)	STOP Condition Setup Time	0.6	—	—	µs
t(HD;DAT)	Data Hold Time	0	—	900	ns
t(SU;DAT)	Data Setup Time	100	—	—	ns
t(LOW)	I²C Clock Low Period	1.3	—	—	µs
t(HIGH)	I²C Clock High Period	0.6	—	—	µs
t(R)	SDA/SCL Rise Time	—	—	300	ns
t(F)	SDA/SCL Fall Time	—	—	300	ns

Note: The DevLab module accepts a **3.3 V to 5.5 V** input supply. An onboard **AP2112K** LDO regulates the voltage to **3.3 V** for the VEML3328 sensor, while a **BSS138 bidirectional level shifter** enables compatibility with both **3.3 V** and **5 V** I²C host systems.

3 Functional Overview

The **UNIT I²C VEML3328 Color Sensor** is a high-performance **RGBIR (Red, Green, Blue, Clear, and Infrared) light-to-digital converter** designed for color sensing, ambient light measurement, and spectral analysis applications. The module integrates the **VEML3328** color sensor together with an onboard **3.3 V voltage regulator** and a **bidirectional I²C level shifter**, allowing direct interfacing with both **3.3 V** and **5 V** host systems.

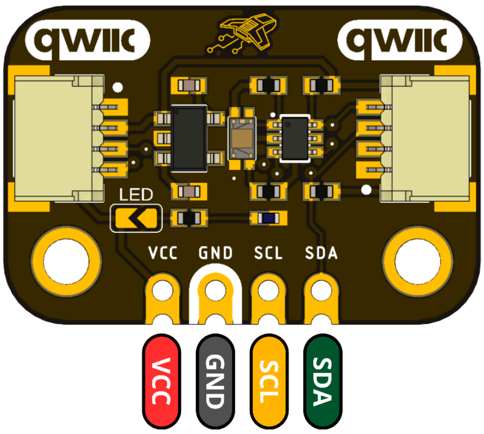
The sensor converts incident light into independent 16-bit digital measurements for the **Red, Green, Blue, Clear, and Infrared** channels, eliminating the need for external analog signal conditioning. Communication is performed through a standard **I²C interface**, accessible from either of the onboard **QWIIC connectors** or the **2.54 mm castellated header**, enabling seamless integration into embedded systems and rapid prototyping platforms.

3.1 I²C Communication Interface

The module operates as an I²C slave device using the fixed **7-bit address 0x10** for configuration and data acquisition. All communication is fully bidirectional through the standard I²C protocol.

Device communication can be verified by reading the **Device ID Register (0x0C)**. The lower byte of this register returns **0x28**, confirming that the sensor is correctly connected and responding on the I²C bus.

The module may be connected using either of the two onboard **QWIIC (JST-SH 1.0 mm)** connectors or through the **2.54 mm castellated header**, providing flexibility for breadboard prototyping and PCB integration.



3.2 Sensor Configuration

The **VEML3328** provides several configurable parameters that allow the user to optimize measurement sensitivity, dynamic range, and conversion time according to the application requirements. All configuration settings are stored in the **Configuration Register (0x00)**, which is a 16-bit register containing multiple user-programmable fields. The most important parameters for normal operation are the **Digital Gain (DG)**, **Analog Gain (GAIN)**, **Integration Time (IT)**, and **Sensitivity**.

The **Digital Gain (DG)** is configured through bits **[13:12]** of the Configuration Register and adjusts the digital amplification applied to the measured signal. The sensor supports three selectable gain levels, with **×1** as the default setting.

Field Value	Digital Gain
00	×1
01	×2
10	×4

The **Analog Gain (GAIN)** is configured through bits **[11:10]** of the Configuration Register. This setting controls the analog front-end sensitivity before analog-to-digital conversion, allowing the sensor to adapt to different illumination conditions. The available analog gain settings are shown below.

Field Value	Analog Gain
11	×½
00	×1
01	×2
10	×4

The **Integration Time (IT)** is configured through bits **[5:4]** of the Configuration Register and determines the duration of each measurement cycle. Increasing the integration time improves sensitivity under low-light conditions, while shorter integration times provide faster measurement updates.

Field Value	Integration Time
00	50 ms
01	100 ms
10	200 ms
11	400 ms

Finally, the **Sensitivity** parameter is controlled by **bit [6]** of the Configuration Register. This setting selects the operating sensitivity of the sensor according to the application requirements.

Field Value	**Sensitivity**
0	**High Sensitivity**
1	**Low Sensitivity**

The measurement results for the Red, Green, Blue, Clear, and Infrared (IR) channels are stored in dedicated 16-bit data registers, which are described in the following section.

3.3 Data Registers

The **VEML3328** stores each color measurement in a dedicated **16-bit read-only register**. Each register is divided into a **Low Byte (LSB)** and a **High Byte (MSB)**. The complete measurement is obtained by combining both bytes into a single 16-bit value.

The register organization is illustrated below.

High Byte	Low Byte
Bits [15:8]	Bits [7:0]

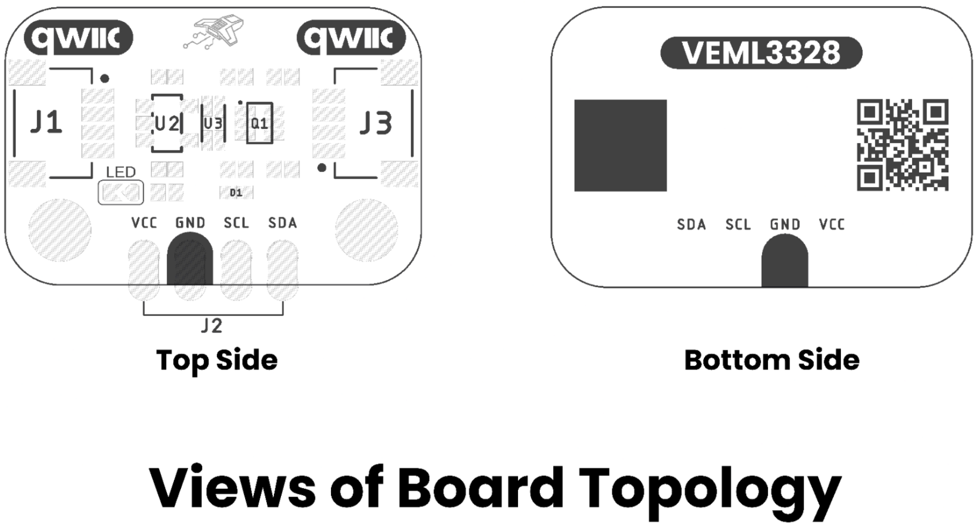
The sensor independently measures the **Red**, **Green**, **Blue**, **Clear**, and **Infrared (IR)** channels. The corresponding data registers are listed in the following table.

Table 3.3.1 Measurement Data Registers

Register	Address	Description
Clear Data	**0x04**	16-bit Clear channel measurement
Red Data	**0x05**	16-bit Red channel measurement
Green Data	**0x06**	16-bit Green channel measurement
Blue Data	**0x07**	16-bit Blue channel measurement
Infrared (IR) Data	**0x08**	16-bit Infrared channel measurement

Finally, the **DevLab VEML3328 Arduino Library** provides high-level functions that automatically read and combine these registers. However, understanding the register organization allows advanced users to configure the device directly through the I²C interface and implement custom applications.

3.2 Board Topology



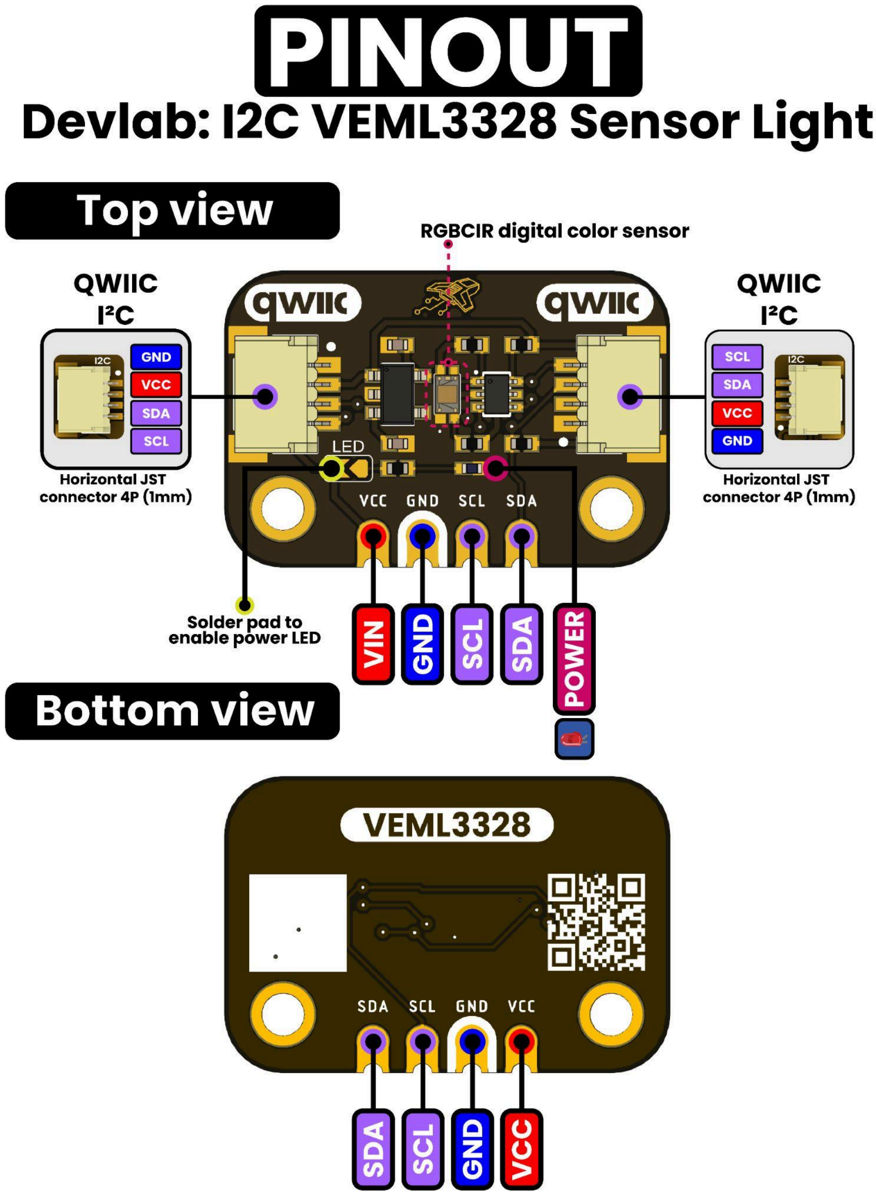
Views of Topology

Table 3.2.1 - Components Overview

Ref.	Description
U1	VEML3328 RGBIR Color Sensor
U2	AP2112K 3.3 V Low Dropout (LDO) Voltage Regulator
Q1	BSS138AKDW Dual Bidirectional I²C Level Shifter
LED(D1)	Power On LED (Red)
J1, J3	QWIIC Connector (JST SH 1 mm pitch) for I²C
J2	2.54 mm Castellated Pin Header

4 Connectors & Pinouts

4.1 General Pinout



DevLab VEML3328 Sensor Module General Pinout

4.2 Pinout General Description

Pin	Label	Description	Typical Use
1	**VCC**	Power supply input for the module. The onboard AP2112K regulator generates the 3.3 V supply required by the VEML3328 sensor.	Connect to a **3.3 V or 5 V** supply from the host microcontroller or development board.
2	**GND**	Ground reference for both power and I²C communication.	Connect to the common system ground.
3	**SCL**	I²C Serial Clock line used to synchronize data transfers between the host and the VEML3328.	Connect to the host I²C **SCL** pin.
4	**SDA**	I²C Serial Data line used for bidirectional communication with the VEML3328.	Connect to the host I²C **SDA** pin.

5 Board Operation

This section demonstrates how to use the **DevLab VEML3328 RGB-IR Color Sensor** with the **DevLab Pulsar C6** using the Arduino framework.

After completing this guide, you will be able to:

- Install the DevLab VEML3328 library.
- Connect the sensor using the I²C interface.
- Read the Red, Green, Blue, IR, and Clear channels.
- Display the sensor measurements in the Serial Monitor.

Requirements

Hardware

- DevLab Pulsar C6
- DevLab VEML3328 RGB-IR Color Sensor
- USB Type-C cable

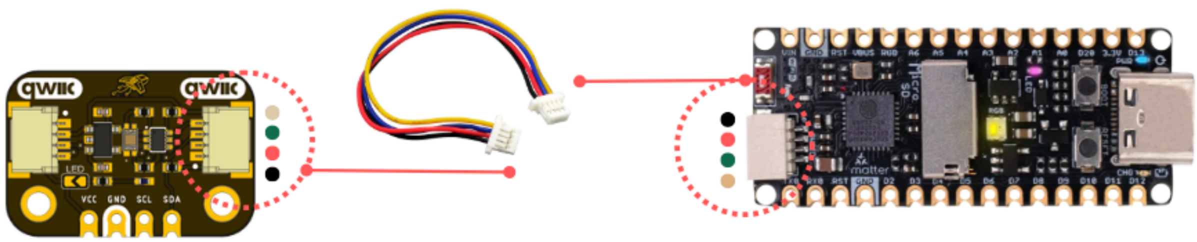
Software

- Arduino IDE
- DevLab VEML3328 Library

Wiring

The sensor communicates using the I²C bus.

VEML3328	Pulsar C6
VCC	3.3 V
GND	GND
SDA	GPIO 6
SCL	GPIO 7



Install the Library

1. Open **Arduino IDE**.
2. Open **Library Manager**.
3. Search for **DevLab VEML3328**.
4. Install the latest version.

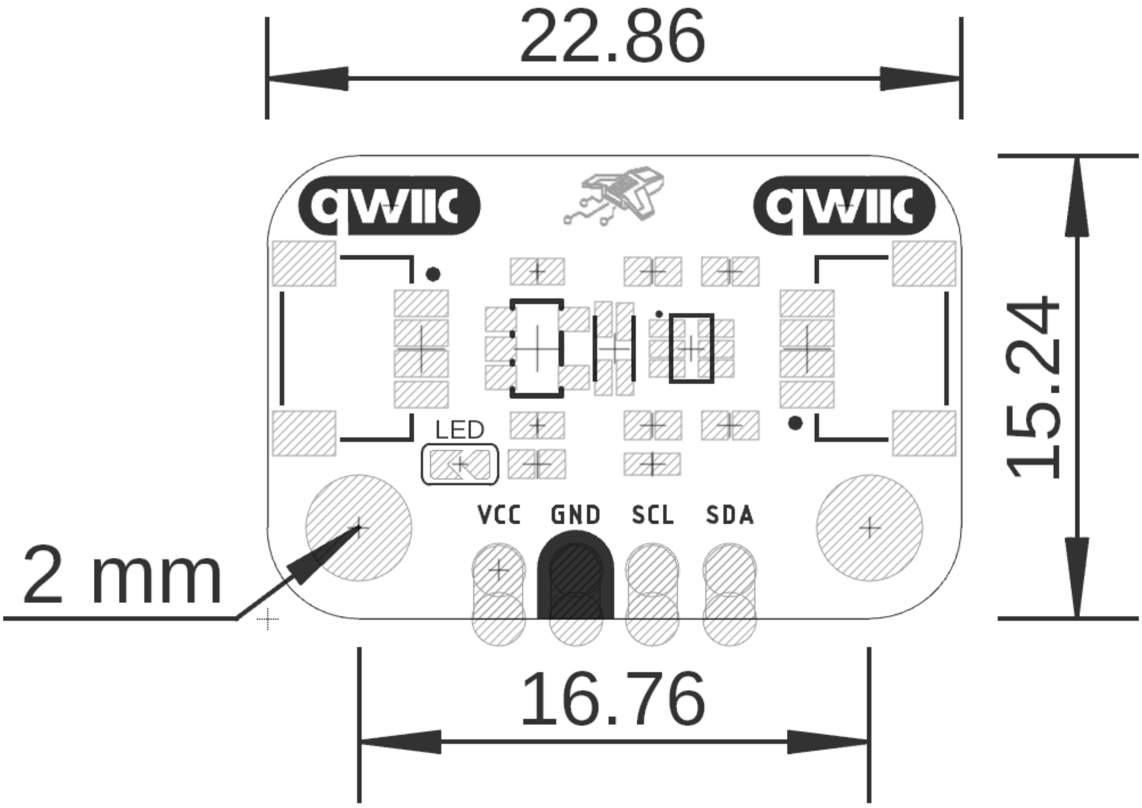
Example Code

Use the **Getting Started** example included with the library or copy the following sketch:

```
#include <Wire.h>
#include <DevLab_VEML3328.h>

#define SDA_PIN 6
#define SCL_PIN 7
#define VEML3328_ADDR 0x10

void setup()
{
   Serial.begin(115200);
   delay(500);
   Wire.begin(SDA_PIN, SCL_PIN);
   Wire.setClock(100000);

   uint8_t estado = DevLab_VEML3328.begin(VEML3328_ADDR, &Wire);

   if (estado != 0)
   {
       Serial.println("Error: VEML3328 not detected");
       while (1)
       {
           delay(1000);
       }
   }
   Serial.println("VEML3328 detected");
   Serial.print("Device ID: 0x");
   Serial.println(DevLab_VEML3328.deviceID(), HEX);
}

void loop()
{
   Serial.println("------------------------");
```

```
  Serial.print("Red    : ");
  Serial.println(DevLab_VEML3328.getRed());

  Serial.print("Green : ");
  Serial.println(DevLab_VEML3328.getGreen());

  Serial.print("Blue   : ");
  Serial.println(DevLab_VEML3328.getBlue());

  Serial.print("IR     : ");
  Serial.println(DevLab_VEML3328.getIR());

  Serial.print("Clear : ");
  Serial.println(DevLab_VEML3328.getClear());

  Serial.println();
  delay(2000);
}
```

6. Mechanical Information



Mechanical dimensions in millimeters

7 Company Information

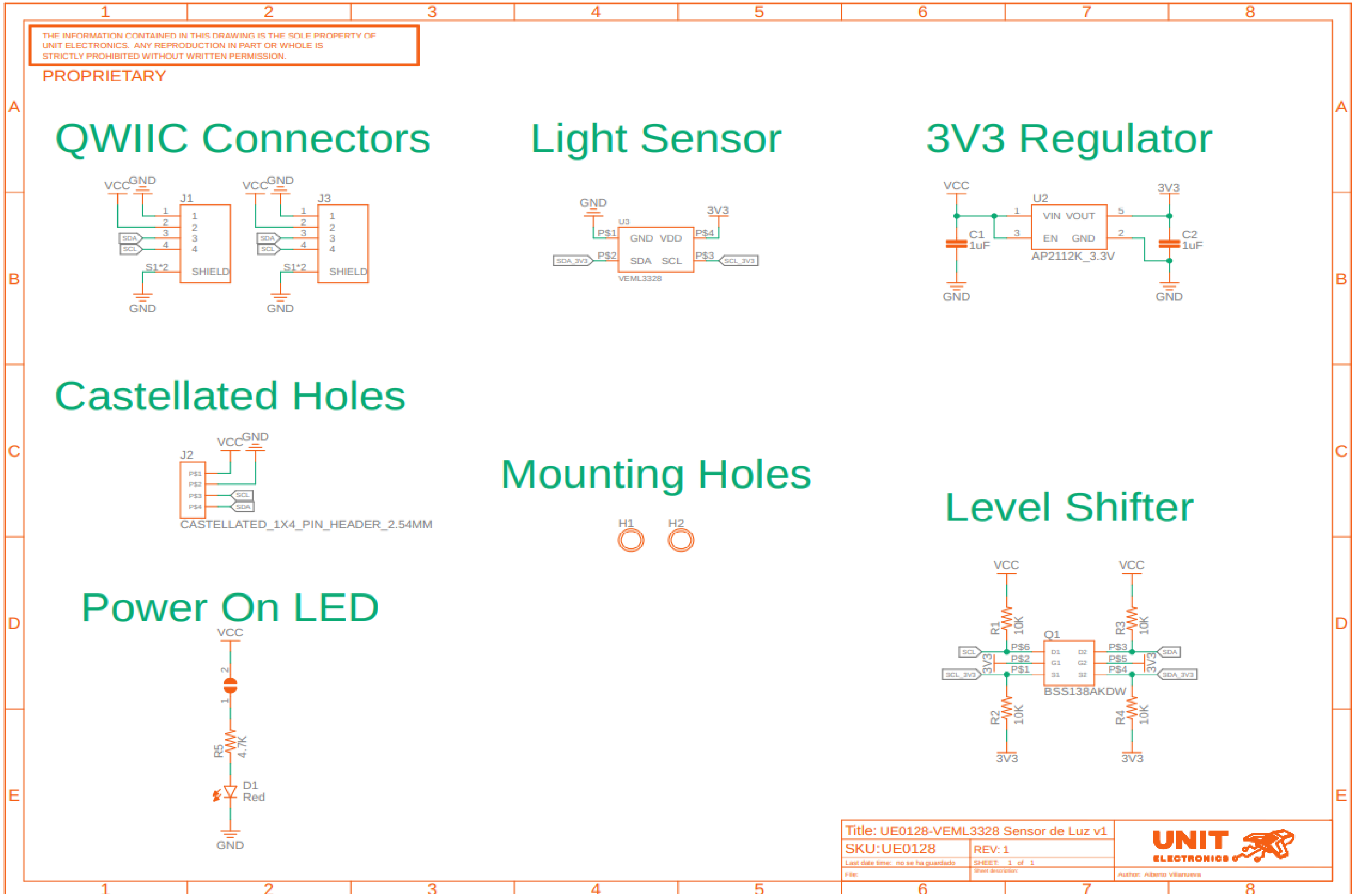
Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	**Link**
Documentation	https://wiki.uelectronics.com/wiki/unit_devlab_i2c_veml3328_color_sensor
Github	https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_veml3328_sensor_light/
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download
Library	https://github.com/UNIT-Electronics-MX/unit_devlab_veml3328_library

9 Appendix

9.1 Schematic
(https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_veml3328_sensor_light/blob/main/hardware/unit_sch_v_1_0_0_ue0128_i2c_veml3328_sensor_light.pdf)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PROPRIETARY
QWIIC Connectors
J1
J3
SHIELD
Light Sensor
U3
GND VDD
SDA SCL
VEML3328
3V3 Regulator
U2
VIN VOUT
EN GND
AP2112K_3.3V
C1 1uF
C2 1uF
Castellated Holes
J2
CASTELLATED_1X4_PIN_HEADER_2.54MM
Mounting Holes
H1
H2
Level Shifter
Q1
BSS138AKDW
Power On LED
R5 4.7K
D1 Red
Title: UE0128-VEML3328 Sensor de Luz v1
SKU:UE0128
REV: 1
UNIT ELECTRONICS